

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices.

The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or

tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then

connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

1. Network Design for a 20-Person Startup:

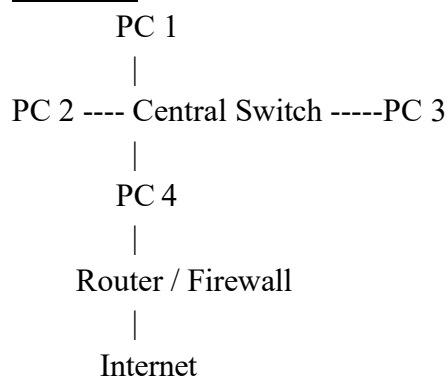
Problem Analysis

A 20-person startup needs a network that is easy to manage, reliable, secure, and expandable. A **star topology** is suitable because every device connects to a central switch.

Proposed Solution

- Use a **24-port managed Layer 2/Layer 3 switch** as the central device.
- Connect computers, printers, and VoIP phones using **Cat6 UTP cables**.
- Connect the switch to a **router/firewall** for internet access.
- Use DHCP for automatic IP address assignment.
- Install a **Wi-Fi access point** for laptops and mobile devices.
- Keep some unused switch ports for future expansion.

Topology



Advantages

1. Easy to manage and troubleshoot.
2. Failure of one cable affects only one device.
3. Easy to add new computers.
4. Centralized network management.
5. Cat6 supports Gigabit Ethernet up to 100 m in the stated design.

The major advantage is **fault isolation**: if one workstation or cable fails, the remaining users can continue working.

Conclusion

Star topology is the best choice for a small startup because it provides simple management, good reliability, and easy expansion.

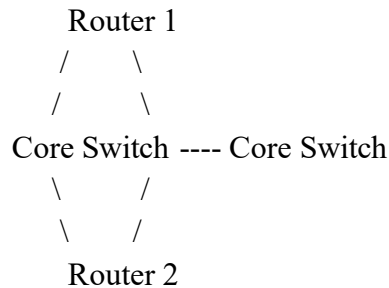
2. Network Design for a Bank:

Problem Analysis

A bank requires **very high availability and minimum downtime** because network failure can affect financial transactions.

Proposed Solution

Use a **full or partial mesh topology** for the core network.



Technical Implementation

- Use multiple core switches.
- Connect important core devices using redundant links.
- Use two enterprise routers for redundant internet connectivity.
- Use **fiber-optic cables** between core devices.
- Use OSPF or BGP for routing.
- Provide redundant power supplies and UPS systems.
- Use backup storage such as RAID.

Advantages

1. Provides multiple communication paths.
2. Failure of one link does not stop the network.
3. Traffic can use an alternative path.
4. High reliability and availability.
5. Suitable for critical banking operations.

The PDF specifically recommends mesh redundancy and alternative routing paths to maintain uptime.

Conclusion

A **partial mesh at the core layer** is a practical choice for a bank because it provides redundancy without the full cost and complexity of connecting every device to every other device.

3. Network Design for a Small Classroom Lab

Problem Analysis

A small classroom lab has a **limited budget** and relatively light network usage. Therefore, a low-cost topology is required.

Proposed Solution

A **bus topology** can be used because computers share a common communication path.

PC 1 ---- PC 2 ---- PC 3----- PC 4
Common Bus Cable

Technical Implementation

- Use a common communication cable.
- Traditional implementation can use **RG-58 coaxial cable**.
- T-connectors and BNC connectors can connect computers.
- 50-ohm terminators are placed at both ends in the classic coaxial design.
- A modern low-cost alternative can use an inexpensive unmanaged switch with Cat5e UTP cables.

Advantages

1. Low hardware cost.
2. Simple installation.
3. No dedicated server is required.
4. Suitable for light classroom activities.
5. Teacher's computer can share files and resources.

Limitations

1. Cable failure can affect all connected computers.
2. Collisions can occur.
3. Performance decreases with increased traffic.
4. Troubleshooting can be difficult.

These cost and limitation considerations are specifically described in the PDF's classroom scenario.

Conclusion

A **bus topology** is suitable for a small classroom with light usage and a tight budget, although it has reliability and collision limitations.

4. Network Design for a Large University Campus

Problem Analysis

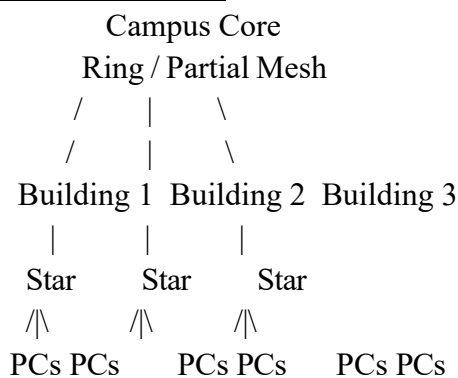
A university has thousands of users distributed across multiple buildings. A single topology is not sufficient, so a **hybrid topology** is appropriate.

Proposed Solution

Combine different topologies at different levels:

- **Ring/Mesh** → Campus backbone
- **Star** → Individual buildings
- **Bus/Tree** → Individual labs

Network Structure



Technical Implementation

- Use high-speed fiber optic cables at the campus backbone.
- Use redundant fiber links between buildings.
- Use distribution switches in each building.
- Use Cat6 cables from access switches to end devices.
- Use the **three-layer architecture**:
 - Core
 - Distribution
 - Access
- Use PoE to connect wireless access points.

Advantages

1. High reliability.
2. Easy to expand.
3. Failure of one backbone link does not isolate a building.
4. Easy management within individual buildings.
5. Supports thousands of users.
6. New buildings can be added without major changes to existing infrastructure.

The PDF describes this hybrid architecture as combining backbone redundancy, building-level simplicity, and campus-wide scalability.

Conclusion

A **hybrid topology** is the best choice for a large university because it combines reliability, scalability, and cost-effective management.

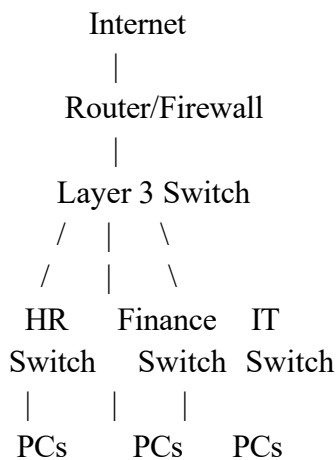
5. Network Design for a Company with Multiple Departments

Problem Analysis

A company with departments such as **HR, Finance, Sales, and IT** needs secure communication while preventing unnecessary broadcast traffic between departments.

Proposed Solution

Use a **hierarchical network** with switches, a Layer 3 switch/router, VLANs, and a firewall.



Technical Implementation

1. Access Layer

Each department gets a dedicated Layer 2 switch.

2. VLANs

Create separate VLANs:

VLAN 10 → HR

VLAN 20 → Finance

VLAN 30 → Sales

VLAN 40 → IT

3. Core Layer

A Layer 3 switch performs **inter-VLAN routing** so departments can communicate when permitted.

4. Security

A router/firewall controls communication with external networks and protects the internal network.

5. IP Addressing

Use proper subnetting for each department.

Advantages

1. Improved security.
2. Reduced broadcast traffic.
3. Better network performance.
4. Easy management.
5. Easy expansion.
6. Redundant links can improve reliability.

The source specifically identifies VLAN separation, inter-VLAN routing, firewall protection, redundancy, and subnetting as key parts of the design.

Conclusion

A **hierarchical VLAN-based network** is suitable for a multi-department company because it provides security, efficient communication, centralized management, and scalability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing

Documentation & Presentation	5%	Clear, professional, and well- documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution